

ESE5180 Lab 1

Wireless Communications

Team 15

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Repository: github.com/Alex-lei-1/ese5180f26-lab1-t15

Submission summary

Part	Key result	Status
1.1 Range	478 m; -99 dBm RSSI; -6 dB SNR	Passed
1.2 Bandwidth	4 ms airtime; 62.5 kbps coded rate	Passed
1.3 Energy	39.20 mA low; 157.89 mA high	Passed
2 LoRaWAN	Decoded JSON delivered through TTN	Passed

All firmware built for **nucleo_wl55jc/stm32wl55xx** with Zephyr 4.4.99. Credentials remain outside version control.

1.1 LoRa Range Challenge

Result	Value
Longest successful distance	478 m
RSSI at longest distance	-99 dBm
SNR at longest distance	-6 dB
Packet airtime	857 ms
Required quiet period	25,710 ms

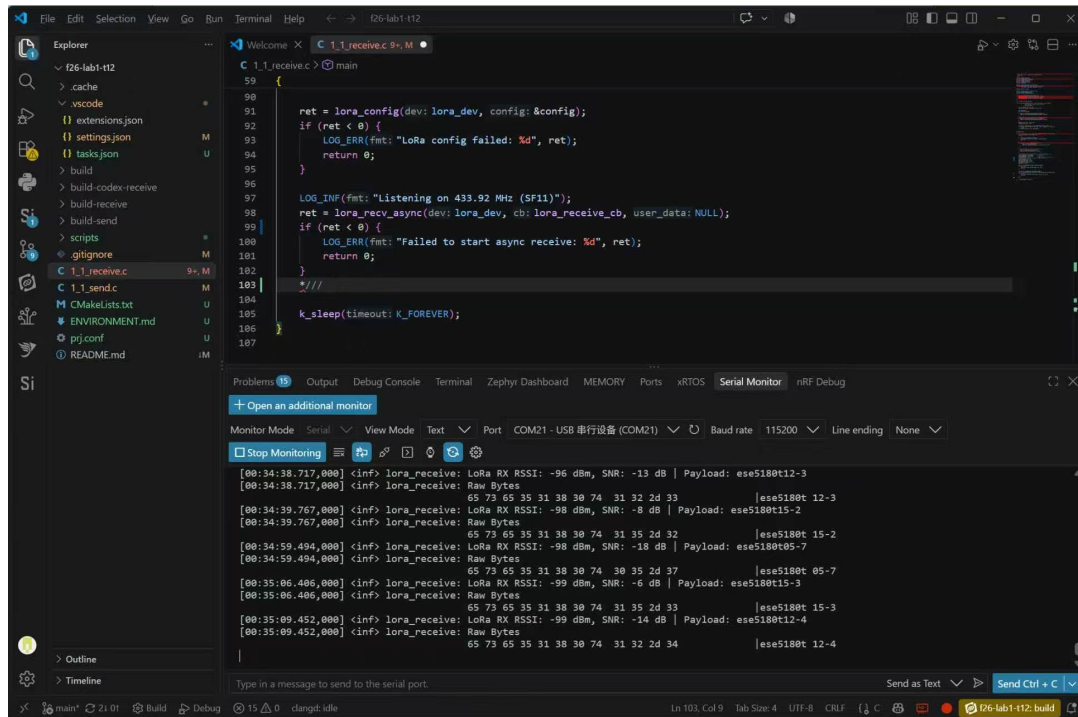


Figure 1. Team 15 packets in the range-test serial capture.

Frequency	Power	BW	SF	CR	Preamble
433.92 MHz	-10 dBm	125 kHz	SF11	4/8	16

Range choices and tradeoffs. High processing gain, stronger coding, a longer preamble, line of sight, and good antenna placement improved range. Costs include low data rate, long airtime, higher energy per packet, and greater channel occupancy. A product should use only the link margin it needs.

Interference. Raw LoRa accepts any packet with matching physical settings. The receiver validates the ese5180t15- prefix and counter to reject other teams' traffic.

1.2 Trading Range for Bandwidth

Parameter	1.1 range	1.2 bandwidth
Bandwidth	125 kHz	500 kHz
Spreading factor	SF11	SF5
Coding rate	4/8	4/5
Preamble	16 symbols	12 symbols
12-byte airtime	857 ms	4 ms
Nominal coded bit rate	0.336 kbps	62.5 kbps

Frequency remained 433.92 MHz and TX power remained -10 dBm. BW500, SF5, CR4/5, and the minimum valid SF5 preamble produced the fastest measured packet airtime: **4 ms**. The coded-rate calculation includes the 4/5 coding factor; 78.125 kbps is the uncoded modulation rate.

```
RX RSSI: -54 dBm | SNR: 9 dB | Payload: ese5180t15-1
RX RSSI: -52 dBm | SNR: 9 dB | Payload: ese5180t15-2
```

Spreading-factor tradeoff

Higher SF increases symbol time, sensitivity, range, airtime, and energy while reducing data rate. Lower SF provides higher throughput and lower packet energy, but less range and link margin.

Suitable low-bandwidth applications

Environmental and agricultural sensors, utility meters, and remote asset trackers send small, infrequent reports. For these systems, long range and battery life matter more than throughput.

Files: 1_2_lora_bandwidth/1_2_send.c and 1_2_receive.c

1.3 Energy Check

Profile	TX power	Peak TX current
Low	-9 dBm	39.20 mA
High	+22 dBm	157.89 mA

Both builds used 915 MHz, BW125, SF10, CR4/5, an 8-symbol preamble, a 12-byte payload, and a 2-second interval. Only TX power changed.

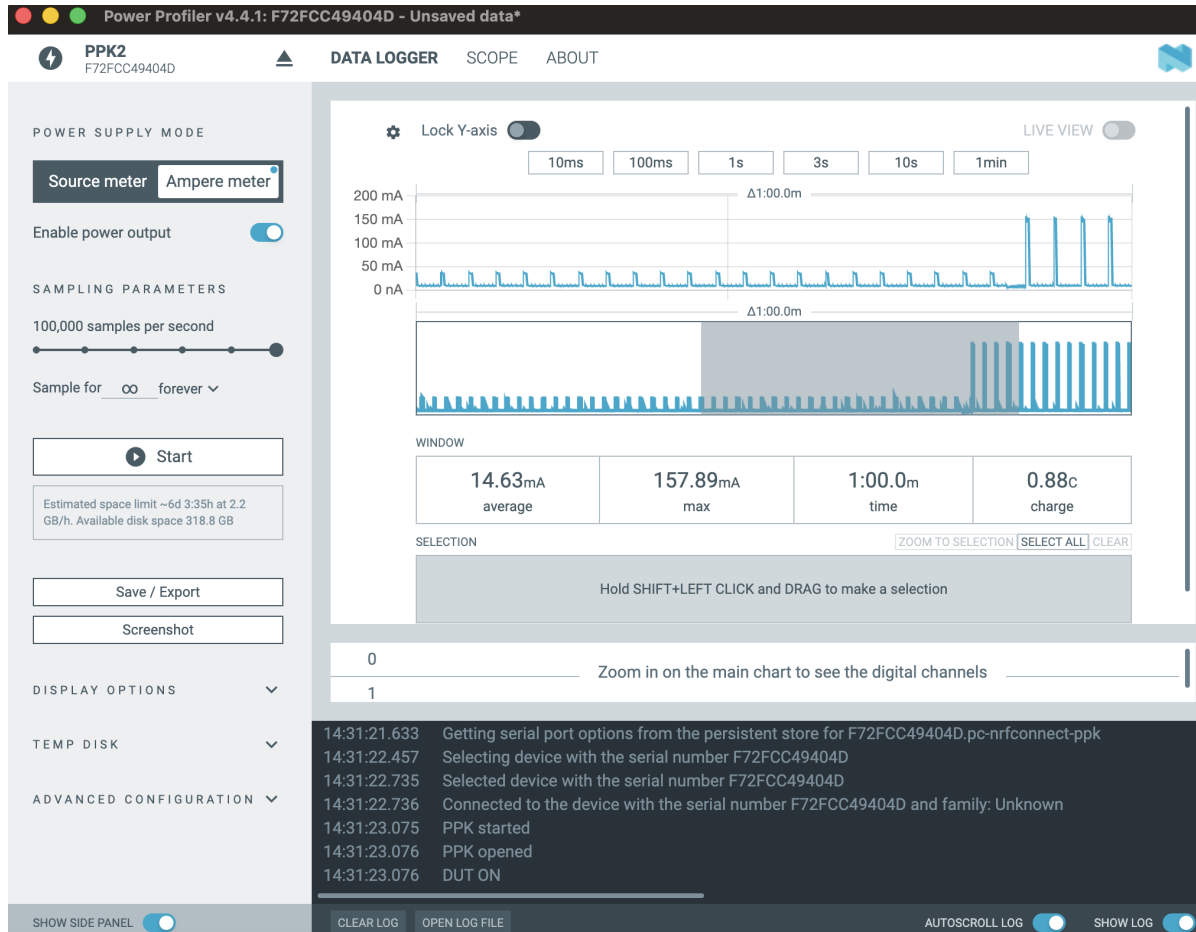


Figure 2. One continuous PPK2 recording showing the low-to-high TX-current transition.

Battery implications. Airtime, message frequency, retries, receive windows, MCU sleep current, peripherals, regulator efficiency, and self-discharge all affect battery life. A coin cell is generally unsuitable for the 157.89 mA pulse because of internal resistance and voltage sag. AA cells or a suitable LiPo can support higher pulses, subject to datasheet limits.

2. LoRaWAN

```
{"name": "Alex", "team": "15", "board": "WL55JC"}
```

Setting	Value
Activation	ABP
Region / channels	US915 FSB2 (8-15 and 65)
Data rate	DR3 / SF7BW125
Class / port	Class A / FPort 2
Uplink	Unconfirmed every 10 seconds
Payload	44 bytes; no trailing null
Observed TTN RSSI / SNR	-88 dBm / 8.75 dB

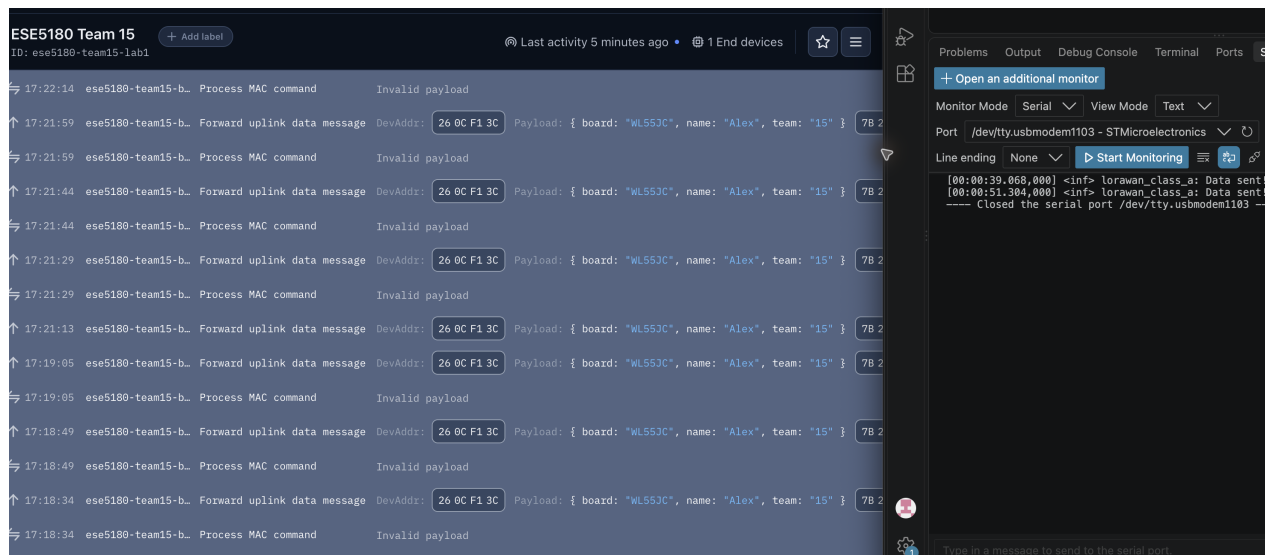


Figure 3. TTN decoded Team 15 JSON while the serial monitor reported successful sends.

Build, flash, serial transmission, gateway reception, and decoded JSON all passed. Credentials remain in gitignored `src/secrets.h`. Removing the sample's `LinkCheckReq` eliminated misleading RX2 timeout messages without affecting unconfirmed uplinks.

LoRaWAN Questions

Confirmed vs. unconfirmed

Confirmed uplinks request an acknowledgement and fit alarms or critical state changes, but acknowledgements and retries cost airtime, latency, and energy. Unconfirmed uplinks fit replaceable periodic telemetry.

Reset session and MAC state

An ABP device and TTN must agree on frame counters, keys, and MAC state. Reflashing can reset device counters while TTN keeps old values, causing replay rejection; reset resynchronizes the lab prototype.

Identifiers and session keys

Raw LoRa provides radio transport only. LoRaWAN uses DevEUI and DevAddr for identity and routing, NwkSKey for network integrity, and AppSKey for application confidentiality, allowing many authenticated devices to share gateways.

Why OTAA is preferred

OTAA performs a join and derives fresh session keys. It improves provisioning, key rotation, device replacement, and network migration, making it safer than fixed ABP credentials for real deployments.

Verification

Component	Evidence
Raw LoRa range	478 m Team 15 packet received
High-throughput LoRa	4 ms packet; consecutive payloads received
Energy	PPK2 transition and peak-current measurements
LoRaWAN	TTN live data, decoded JSON, and serial output
Security	No DevEUI, DevAddr, AppSKey, or NwkSKey committed

All required source files, results, explanations, and screenshots are included in the Team 15 repository.